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HW4 CHE 321

DUE 18/FEB/2020

100

1) a liquid phase $A+B \rightleftharpoons C$ ($A_0 = B_0 = 2M$ $K_c = 10 L/mol$)

$$K_c = \frac{C_c}{C_A C_B} \quad (A = A_0(1-x) \quad C = A_0(x)$$

$$K_c [A_0(1-x)]^2 = A_0 x$$

$$K_c (A_0 (1-x)^2) = x$$

$$K_c (A_0) = \frac{x}{(1-x)^2} = 10 \cdot 2 = 20$$

$$x = .80 \text{ good seek}$$

$$(A=B = A_0(1-x) = 2 \cdot .2 = 0.4M$$

$$C = A_0(x) = 2 \cdot .8 = 1.6M$$

B) gas $A \rightarrow 3C$ $\Delta P = 0$ $T = 400K$ $P = 10 \text{ atm}$ $K_c = .25$

$$\epsilon = y_{A0} \delta \quad y_{A0} = 1 \quad \delta = 2$$

$$K_c = \frac{C_c^3}{C_A} \quad (A = \frac{F_{A0}}{V} = \frac{F_{A0}(1-x)}{V_0(1+\epsilon x)} = \frac{A_0(1-x)}{(1+2x)}$$

$$C_{A0} = \frac{PV}{nRT} \quad (C = \frac{F_{A0}(3x)}{V_0(1+2x)} = \frac{A_0 3x}{(1+2x)}$$

$$C = \frac{P}{RT}$$

$$C = \frac{10}{400 \cdot .082057} = .305 \quad K_c = \frac{F_{A0}(3x)^3}{(A_0(1-x)(1+2x))^3}$$

$$K_c = .25 \quad C_{A0} = .305 \quad x = .58 \text{ solver in excel}$$

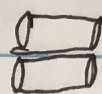
$$A = .0593 \text{ M}$$

$$C = .25M$$

2
 $965 \text{ A} \rightarrow \text{B}$ $k = .0015 \text{ min}^{-1}$, $T = 80^\circ\text{F}$ Pipe 10 ft $D = 1 \text{ in}$

$x = .9$ $E_A = 25,000 \text{ cal/mol}$ MM 58

parallel PFR's find # of tubes

 $\rightarrow F_B = 1000 = F_{A0}(x)$

$$\frac{dF}{dV} = r_A \quad r_A = -k(A) = -k(A_0(1-x))$$

$$V_t = F_{A0} \int_0^x \frac{dx}{-r_A} = \frac{F_{A0}}{k(A_0)} \int_0^x \frac{1}{1-x} dx \rightarrow -\ln(x-1) \Big|_0^x$$

$$V_t = \frac{F_{A0}}{k(A_0)} - \ln(1-x)$$

$$F_{A0} = \frac{F_B}{x} = \frac{1000}{.9} = 1111.1 \text{ lb/hr}$$

K from $k = A e^{-E_A/RT}$

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_A}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$\ln\left(\frac{k_2}{.0015}\right) = \frac{25,000 \text{ cal/mol}}{1.985 \text{ cal}} \left(\frac{1}{299.67 \text{ K}} - \frac{1}{399.67 \text{ K}}\right)$$

$$k_2 = 55.33 \text{ solver excell} = 3319.83/\text{hr}$$

$$V_{tot} = -\ln(1-x) \frac{F_{A0}}{k(A_0)}$$

$(A_0 = 7) \quad PV = nRT \quad (= P/RT)$

$$P = (RT) = 132 \text{ psia} + 14.7$$

$$F_{A0} [=] \text{ lb/hr} \quad \frac{\text{lb}}{\text{hr}} \left(\frac{453.6 \text{ g}}{1 \text{ lb}} \right) \left(\frac{1 \text{ mol}}{58 \text{ g}} \right)$$

$$(A_0 = .0189 \text{ lbmol/ft}^3) \quad 10.73 \cdot 719.67 \text{ R}$$

$$F_{A0} = 1111.1 \times 453.6 \div 58 = 8689.6 \text{ mol/hr}$$

$$(A_0 [=] \text{ lbmol/ft}^3) \left(\frac{453.6 \text{ g}}{1 \text{ lb}} \right) = 8.57 \text{ mol/ft}^3$$

$$V_{tot} = -\ln(1-x) \cdot \frac{8689.6}{(3319.83)(8.57)} \text{ mol/hr ft}^3 = .703 \text{ ft}^3$$

$$1 \text{ tube} = \pi \left(\frac{1 \text{ ft}}{24} \right)^2 \cdot 10 \text{ ft} = .0545 \text{ ft}^3$$

$$\# = \frac{.703 \text{ ft}^3}{.0545 \text{ ft}^3} = 12.9 \text{ so } \boxed{13 \text{ tubes}}$$